Logic definitions

1

a The process of reasoning to a general conclusion through observation of specific cases.

b The process of reasoning to a specific conclusion from a general statement.

c A sentence that can be judged either true or false.

d A statement that conveys only one idea.

e A statement that consists of two or more simple statements.

2

a Yes **b** Yes **c** No **d** Yes

3 Example answer: Soup, pasta, and carrots.

Compound statements and negation

4

a

i Compound statement **ii** Conditional ($\Longrightarrow$)

b

i Simple statement

c

i Compound statement **ii** Biconditional ($\Longleftrightarrow$)

d

i Compound statement **ii** Negation ($\sim$)

e

i Compound statement **ii** Conjunction ($\wedge$)

f

i Compound statement **ii** Conjunction ($\wedge$)

g

i Compound statement **ii** Conditional ($\Longrightarrow$)

h

i Simple statement

i

i Compound statement **ii** Biconditional ($\Longleftrightarrow$)

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5

a Some humans are not able to swim.

b Some houses are not connected to the internet.

c All birds have flippers.

d Some Libras are Capricorns.

e Some tricycles have two wheels.

f Some cats do not have feet.

g No cars are driving across the bridge.

h All doctors are paediatricians.

i Some skydivers are scientists.

j No biscuits contain sugar.

Symbolic form

6

a $\sim$ **b** $\wedge$ **c** $\vee$ **d** $\implies$

e $\iff$

7

a $p \wedge \sim q$ **b** Conjunction

8

a $p \implies (q \vee r)$ **b** Conditional

9

a $\sim (p \implies \sim q)$ **b** Negation

10

a $(p \implies q) \vee \sim r$ **b** Disjunction

11

a $(p \vee q) \implies r$ **b** Conditional

12

a $(p \implies q) \iff \sim r$ **b** Biconditional

13

a $p \iff (\sim q \vee r)$ **b** Biconditional

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14

a $(p \wedge q) \implies r$ **b** Conditional

15

a $(p \iff q) \vee r$ **b** Disjunction

16

a $\sim p$ **b** $p \wedge q$ **c** $\sim q \vee \sim p$ **d** $\sim q \iff \sim p$

e $\sim p \implies \sim q$ **f** $\sim q \wedge p$

17

a $\sim q \iff p$ **b** $p \implies \sim q$ **c** $\sim p \wedge q$ **d** $\sim p \wedge \sim q$

e $\sim (p \vee q)$ **f** $\sim (\sim q \implies p)$

18

a James does not have a smartphone.

b Dave has a laptop and James has a smartphone.

c If Dave does not have a laptop, then James has a smartphone.

d Dave does not have a laptop or James does not have a smartphone.

e It is false that Dave has a laptop and James has a smartphone.

19

a Quentin does not have a rabbit.

b Quentin has a rabbit or Maximilian has a dog.

c Quentin does not have a rabbit if and only if Maximilian does not have a dog.

d It is false that Quentin has a rabbit or Maximilian has a dog.

e Quentin does not have a rabbit and Maximilian does not have a dog.

20

a $(p \wedge \sim q) \wedge r$ **b** $(\sim p \wedge q) \wedge r$ **c** $(p \wedge q) \vee r$ **d** $p \implies (q \vee \sim r)$

21

a $(r \wedge q) \implies p$ **b** $(\sim p \iff \sim q) \vee \sim r$

c $(r \iff q) \wedge p$ **d** $q \implies (p \iff r)$

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22

a The water is $28^\circ\mathrm{C}$ or the sun is down, and we do not go swimming.

b The water is not $28^\circ\mathrm{C}$, and the sun is down or we go swimming.

c If we do not go swimming, then the sun is down and the water is $28^\circ\mathrm{C}$.

d If the sun is down then we go swimming, and the water is $28^\circ\mathrm{C}$.

e The sun is down if and only if the water is $28^\circ\mathrm{C}$, and we go swimming.

23

a The water is $60^\circ\mathrm{F}$ and the sun is up, or we go snorkelling.

b If the sun is up then the water is $60^\circ\mathrm{F}$, or we go snorkelling.

c If the sun is up and we go snorkelling, then the water is $60^\circ\mathrm{F}$.

d If the water is not $60^\circ\mathrm{F}$, then the sun is up or we go snorkelling.

e If the sun is up, then the water is $60^\circ\mathrm{F}$ if and only if we go snorkelling.

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